Example 3.22

Factorise $49x^2 - 84xy + 36y^2$

Solution:

Now, $49x^2 - 84xy + 36y^2$

$$7^2x^2 - 84xy + 6^2y^2 = (7x)^2 - 84xy + (6y)^2$$

Comparing this with $a^2 - 2ab + b^2 = (a-b)^2$ we get $a = 7x,\ b = 6y$

$$(7x)^2 - 2(7x)(6y) + (6y)^2 = (7x-6y)^2$$

$$\therefore\ 49x^2 - 84xy + 36y^2 = (7x-6y)^2$$

$(7x–6y)$, $(7x–6y)$ are the two factors.

Try these

Find the factors

factor 1	factor 2	product	sum
		35	12
		-40	-3
		60	-17
		-51	+14
		-32	-4

Example 3.23

Factorise : $49x^2 - 64y^2$

Solution:

Now, $49x^2 - 64y^2$

$7^2x^2 - 8^2y^2 = (7x)^2 - (8y)^2$

Comparing this with $a^2 - b^2 = (a+b)(a-b)$ we get $a = 7x,\ b = 8y$

$$(7x)^2 - (8y)^2 = (7x+8y)(7x-8y)$$

$(7x+8y)$, $(7x–8y)$ are the two factors.

Type 5 : Factorisation of the expression ($ax^2 + bx + c$)

Example 3.24

Factorise $x^2 + 8x + 15$

Product = 15	Sum = 8
1×15 = 15	1+15 = 16
3 × 5 = 15	**3+5 = 8** ✓

Solution:

Given $x^2 + 8x + 15$

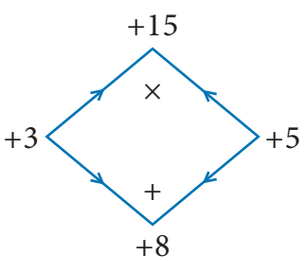
This is in the form of $ax^2 + bx + c$

We get $a = 1, b = 8, c = 15$

Now, the product $= a \times c$ and sum $= b$

$= 1 \times 15$ $\quad b = 8$

$= 15$



$= x^2 + 8x + 15$

$= x^2 + 3x + 5x + 15$ (the middle term $8x$ can be written as $3x+5x$)

$= (x^2 + 3x) + (5x + 15)$

$= x(x+3) + 5(x+3)$ taking out the common factor $x+3$)

$x^2 + 8x + 15 = (x+3)(x+5)$

Therefore, $(x+3)$, $(x+5)$ are the two factors.

Think

$x^2–4(x–2) = (x^2–4)(x–2)$

Is this correct? If not correct it.

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